

Programme	B. Sc. Computer Science				
Course Title	Computer Essentials with Word Processing & Presentation				
Type of Course	Minor				
Semester	I				
Academic Level	100-199				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	1. Fundamental Mathematics Concepts: Number System				
Course Summary	This course serves as an introductory exploration into the foundational concepts of computing. Through a combination of lectures, hands-on exercises, and practical assignments, participants develop a holistic understanding of computer fundamentals. Ultimately, this course serves as a cornerstone for further studies in computer science, information technology, and related disciplines, empowering learners to navigate and contribute to the ever-evolving landscape of computing.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Gain proficiency in understanding and representing data in various forms, including binary, decimal, hexadecimal, and character encodings.	Ap	F	Instructor-created exams / Quiz
CO2	Understand the basic principles of computer architecture and organization	U	C	Assignment / Demonstrations
CO3	Understand the concept of software and its significance in computing and be familiar with various types of software, including system software, application	U	C	Seminar Presentation / Group Tutorial Work

	software, and utility software.			
CO4	Understand the basic principles of document design and layout for enhanced readability and visual appeal	Ap	P	Hands-on practical sessions
CO5	Understand the importance of effective communication and visual aids in presentations.	Ap	P	Hands-on practical sessions
CO6	Acquire practical skills through hands-on exercises and projects, preparing participants to apply their knowledge in academic, professional, and personal contexts.	Ap	P	Hands-on practical sessions
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs	Marks
I	Introduction to Computer Science and data representation		10	20
	1	Introduction to Computers: Generation, Classification, Characteristics of Computers, Significance	2	
	2	Number Systems :Binary, Decimal, Octal, Hexadecimal.	2	
	3	Conversion from one base to another	3	
	4	Computer Codes: BCD Code, Excess 3 Code, ASCII Code, Unicode, Gray Code	3	
II	Basic Computer Organization		10	20
	5	CPU organisation :Arithmetic and Logic Unit, Control Unit	1	
	6	Memory hierarchy: Registers, Cache, Primary Memory, Secondary Memory	2	
	7	Primary Storage: RAM(SRAM, DRAM), ROM(Masked ROM , PROM, EPROM, EEPROM)	2	
	8	Secondary storage: SSD, HDD, Magnetic tapes, Disk Storage	2	

	9	Input/Output Unit:- Input Device: Keyboard, MouseTouchpad, Trackball, Scanner, Graphics Tablet, Microphone, Webcam, Joystick/Gamepad, Biometric Input Devices Output Devices: Monitor/Display, Printer,Projector, Speakers, Headphones, Plotter	3	
III	Understanding Softwares		10	20
	10	Introduction to Software (Definition and Importance of Software, Types of Software-System software, Application Software, Prop oratory vs Open source)	2	
	11	Operating Systems (Introduction to Operating Systems , Common Operating Systems,User Interfaces)	2	
	12	Device Drivers and Utilities (Device Drivers , System Utilities ,Productivity Software ,Multimedia Software)	2	
	13	Computer languages(Machine, Assembly and HighLevel), Language Translator- Assembler, Compiler, Interpreter	2	
	14	Security Software and Best Practices(Antivirus Programs ,Firewalls and Security Suites,Best Practices for Software Security)	2	
IV	Introduction to Word Processing & Presentation		15	10
	15	Basics of Word Processing: Creating, Opening, Saving, and Closing Documents, Text Entry and Formatting (Font, Size, Color),Paragraph Formatting (Alignment, Spacing),Copying, Cutting, and Pasting Text,Spell Check and Grammar Check)	2	
	16	Advanced Word Processing Techniques (Styles and Templates,Tables and Graphics (Inserting, Formatting),Headers and Footers,Page Layout (Margins, Orientation),Document Views (Print Layout, Draft, Outline	2	
	17	Advanced Graphics and Multimedia(SmartArt and Shapes , Customizing SmartArt and shapes,Embedding and Linking Media,Advanced techniques for embedding and linking images, audio, and video)	2	
	18	),Document Collaboration (Track Changes, Comments),Mail Merge for Personalized Documents)	1	
	19	Introduction to Presentation Software (Creating a New Presentation, Slide Basics (Adding, Deleting, Rearranging), Slide Layouts and Choosing Templates,Text Entry and Formatting Inserting and Formatting Images and Shapes)	2	

	20	Enhancing Presentations with Multimedia (Inserting and Formatting Media (Audio, Video),Transitions Between Slides,Master Slides for Consistent Formatting,Design and Themes for Visual Appeal)	3	
	21	Animations for Text and Objects (Slide Show Setup (Timings, Rehearsal)	2	
	22	Effective Presentation Delivery (Tips for Engaging Presentations, Presenter View and Speaker Notes, Handling Q&A Sessions, Dealing with Technical Issues, Customizing Presentations for Different Audiences, Printing and Exporting Slides	1	
V	Hands-on Word Processor and Presentation Tool:		30	
	Practical Applications, Case Study and Course Project			
		Identification and familiarization of Hardware Components (Processor, RAM,ROM, Peripheral devices, SSD, HDD, SMPS, Motherboard, Ports)	5	
		Microsoft Word: - Document Formatting: - Create a new document, set margins to 1 inch, and change the page orientation to landscape. - Apply a consistent font style, size, and color to the entire document. - Paragraph Formatting: - Create a bulleted or numbered list with at least three items. - Adjust the indentation and line spacing for a specific paragraph. - Headers and Footers: - Insert a header with the document title and page number on the right. - Add a footer with the date aligned to the center. - Tables and Graphics: - Create a table with four columns and three rows. - Insert an image into the document and adjust its position. - Styles and Themes: - Apply a heading style to a section of text. - Change the document theme to give it a different look. Microsoft PowerPoint:	20	

		6. Slide Creation: ● Create a new PowerPoint presentation and add five slides. ● Apply different slide layouts to each slide. 7. Text and Object Formatting: ● Add a title to the first slide and format it with a unique font and color. ● Insert a shape and customize its fill and outline colors. 8. Transitions and Animations: ● Apply a slide transition between the first and second slides. ● Add an entrance animation to a text box on the third slide. 9. Master Slides: ● Customize the master slide with a background color or image. ● Add a placeholder for slide numbers in the master slide. 10. Delivery and Export: ● Set up presenter view for a slideshow. Export the presentation as a PDF document		
		Case Study: Exploring feature of PowerPoint to enhance presentation skill	5	

Reference Books:

1. Tanenbaum, Andrew S. and Herbert Bos. Modern Operating Systems. 4th ed., Pearson, 2014.
2. Brookshear, J. Glenn. Computer Science: An Overview. 13th ed., Pearson, 2014.
3. Norton, Peter. Introduction to Computers. 7th ed., McGraw-Hill, 2016.
4. Patterson, David A. and John L. Hennessy. Computer Organization and Design: The Hardware/Software Interface. 5th ed., Morgan Kaufmann, 2013.
5. Stallings, William. Computer Organization and Architecture: Designing for Performance. 10th ed., Pearson, 2016.
6. Hennessey, John L. and David A. Patterson. Computer Architecture: A Quantitative Approach. 6th ed., Morgan Kaufmann, 2017.

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PS O5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	3	-	-	-	-	-						
CO 2	2	3	-	-	-	-						

CO 3	-	-	1	1	-	-						
CO 4	-	-	2	3	-	-						
CO 5	-		-	3	-	-						
CO 6	-	-	-	3	-	-						

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4			✓	✓
CO 5			✓	✓
CO 6			✓	